

CS349 : Artificial Intelligence-II

End Semester Exam (27/11/2024 : 9:30 AM – 12:30 PM)

Maximum time: 3 Hours

Total Marks: 90

Answer all of the questions.

Make reasonable assumptions as and whenever necessary. However, state them clearly. You can answer the questions in any sequence. However, the answers to all the parts of any particular question should appear together. No doubts will be cleared during the exam. No partial marks will be given.

Notations are used as discussed in class, or as per their usual meaning.

No marks will be given without proper explanation/ justification.

1. Consider the following update rule for Gradient Descent (GD). $w_i := w_i - \alpha \nabla_{w_i} \mathcal{J}$ [4 + 4 + 2 = 10]
 - (i) How does $\nabla_{w_i} \mathcal{J}$ help to converge the GD algorithm to minimize the cost?
 - (ii) What is the solution to the *exploding gradient* problem? How does this solution ensure the above convergence?
 - (iii) What will happen for high and low values of α ?
2. Write True/ False with justification. (No marks will be given without proper justification) [4 × 5 = 20]
 - (i) We can introduce non-linearity in ANN (Artificial Neural Network) architecture by introducing more number of hidden layers.
 - (ii) By increasing *dropout rate* in ANN, we can solve *high-variance* problem.
 - (iii) ReLU may help in *overfitting* problem.
 - (iv) *Softmax* is usually preferred as the output function in ANN for regression problem.
 - (v) Increasing *number of filters* in CNN helps to expand the dimension of the input tensor.
3. Write to-the-point: [4 × 5 = 20]
 - (i) Advantage(s) of *tanh* over *sigmoid*
 - (ii) Advantage(s) of *ADAM* over *RMSprop*
 - (iii) Two disadvantages of ReLU
 - (iv) Advantage of MSE over MAE, and Advantage of MAE over MSE
 - (v) Bias-Variance trade off
4. Consider the convolution operations of Figure 1: [11 + 4 + 2 + 2 + 1 = 20]
 - (i) What are the values of $a, b, c, d, e, f, g, h, i, j, k$?

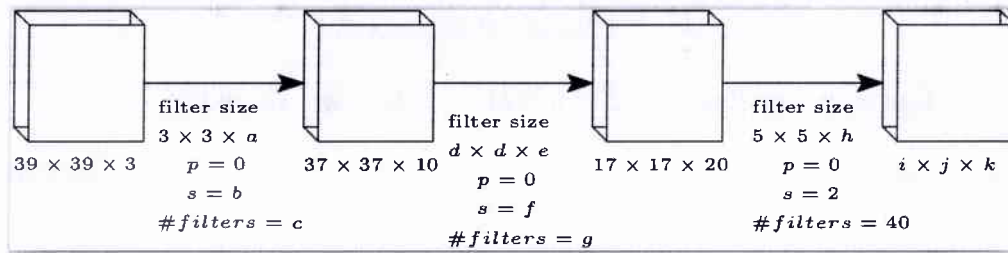


Figure 1: Convolution operations (all the symbols used are of their usual meaning)

- (ii) What is the number of parameters needed to be learned here?
- (iii) Which hyperparameters are used here?
- (iv) If the first layer (i.e., $39 \times 39 \times 3$) is connected with the last layer (i.e., $i \times j \times k$) using a fully connected network, how many parameters must be learned?
- (v) What kind of padded convolution is used here: Valid or Same?

5.

[4 + 2 + 4 = 10]

- (i) Suppose you have n training samples with number of features as k .
The normal equation is $\theta = (X^T X)^{-1} X^T y$.
For the given values of number of training samples (n) and features (k), what are the dimensions of θ , X , and y ?
- (ii) Suppose you train a logistic regression model $h_\theta(x) = \sigma(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$.
Assume $\theta_0 = 5, \theta_1 = -1, \theta_2 = -1$; what is the equation of the decision boundary?
Mention which side of the decision boundary (i.e., the side, where the origin (0, 0) belongs; or the opposite side) you will have the positive class?
- (iii) Suppose you have the following dataset:
(1, 30), (2, 45), (3, 51), (4, 57), (5, 60), (6, 65), (7, 70), (8, 71).
You want to use single-valued linear regression. Using normal equation, write down the equation for h_θ .

6.

[2 + 4 + 4 = 10]

- (i) You have a 10000-word vocabulary, and are using an LSTM with 100-dimensional activations $a^{<t>}$. What is the dimension of the update gate (Γ_u) at each time step?
- (ii) What are the differences between Back-Propagation (BP) and Back-Propagation Through Time (BPTT)?
- (iii) What are the differences between GRU and LSTM?